Relation-Aware Geometric Grounding and Cost-to-Go MPPI for Natural Language Command Execution

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***Abstract*—**We present a modular offline VLA agent for the Meta-Sejong AI Robotics Challenge (MARC) 2026 that couples language-guided CCTV grounding with mobile manipulation. Landmark and area cues constrain camera selection, relation-aware ranking resolves target–anchor candidates, and support-plane projection with ankle correction estimates world coordinates. For retrieval, cost-to-go-guided model predictive path integral (MPPI) control descends a grid navigation function under chassis constraints, and a depth-prioritized occupancy grid lets depth-confirmed floor override LiDAR evidence to suppress persistent ground misclassification. Over seven runs of 33 public missions, the Stage 1 score is 75.8/100 with 87.9% camera accuracy. On five public retrieval missions, the final system completes all 20 runs, and the depth-prioritized grid eliminates path loss in the recorded dense-area trials. Remaining failures involve perception near curbs.

Keywords*—VLA, visual grounding, spatial relations, geometric localization, model predictive path integral control, occupancy grid, mobile manipulation*

I. INTRODUCTION

MARC 2026 couples language-guided visual grounding and object retrieval in an Isaac Sim campus digital twin [1]. Stage 1 requires selecting a CCTV camera and reporting semantic fields and world coordinates; Stage 2 requires retrieving an object and delivering it to its owner. Repeated object classes across largely non-overlapping cameras and imperfect detections make confidence-only selection unreliable, bounding boxes do not identify the physical points required for localization. In navigation, a map that only accumulates LiDAR hits retains spurious returns, and slope-induced ground misclassification repeatedly marks floor cells as obstacles; both can close every free path to the goal.

Our system builds on YOLO11 [2], landmark-based grounding [3], ray–plane geometry [4], and MPPI [5]; posture is interpreted with support context rather than used alone as a danger cue, complementing pose-based fall detection [6]. Our contributions are:

- **Constraint-based camera gating and relation-aware candidate selection** that narrow the cameras without emptying the candidate set
- **Support-aware localization with posture-dependent ankle correction** on the support surface
- **Cost-to-go-guided MPPI**: a Dijkstra cost-to-go field with MPPI under in-rollout chassis limits
- **Depth-prioritized occupancy grid** in which depth-confirmed floor overrides LiDAR log-odds evidence

Evaluation on public missions does not establish generalization to unseen environments.

II. PIPELINE

Fig. 1 shows the system, a modular VLA pipeline in which language constrains vision and the grounded result drives action, rather than an end-to-end policy. A mission string m and frames from N = 6 fixed CCTV cameras enter two independent branches. A rule-based parser maps the mission, via synonyms from the official asset list, to target, landmark, relation, posture, and area cues. A YOLO11 detector covers 34 classes (10 targets, 15 landmarks, and nine postures), and an auxiliary tiled detector handles three small-object classes: sunblock, tissue, and disposable cup.

Stage 1 (Section III) joins the branches in camera gating and relation-aware selection, assesses the situation for person missions, and localizes the target and anchor, producing the GroundingResult $y = (c^*, o, l, r, s, x_t, x_a)$. Stage 2 (Section IV) reuses y: x_t sets the goal of a cost-to-go field built on a depth-prioritized occupancy grid, and MPPI drives the robot to the object and, after a depth-based grasp, to the owner zone.

III. STAGE 1: RELATION-AWARE GROUNDING WITH SUPPORT-AWARE LOCALIZATION

A. Camera Gating and Candidate Selection

A landmark gate retains cameras containing the named landmark l. An area gate uses indicative objects, such as vehicles for parking areas and benches or picnic tables for parks, and keeps the dominant camera:

$$G_L = \{\, c : \exists\, d \in D_c,\ \mathrm{class}(d) \approx l \,\} \tag{1}$$

$$G_A = \{\, \mathrm{argmax}_c\, n_c \,\} \quad \text{if } n^{(1)} \geq \gamma\, n^{(2)} \tag{2}$$

where D_c is the detection set of camera c, n_c its indicator count, and $n^{(1)}$, $n^{(2)}$ the two largest counts. Applicable gates are intersected, with fallback to the landmark gate or all cameras to avoid an empty candidate set. This fallback does not guarantee retention of the correct camera when landmarks are missed.

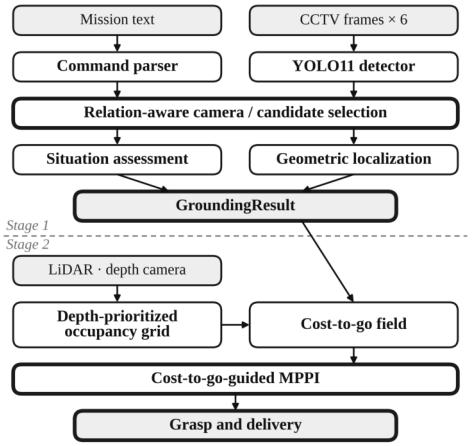


Fig. 1. System pipeline. Stage 1 grounds the mission in the CCTV views; Stage 2 drives to x_t, grasps the object, and delivers it.

Within each camera, targets are paired with their nearest anchors using footprint separation. With g the oriented-box surface gap and δ_{near} the proximity scale, proximity and the cross-camera ranking are

$$\pi = \max(\,0,\, 1 - \max(0, g) \,/\, \delta_{near}\,) \tag{3}$$

$$c^* = \operatorname{argmax}_{c \in G} \kappa_c, \quad \kappa_c = (\,b_c, \pi_c, s_c\,) \tag{4}$$

where b_c is relation agreement, s_c detection confidence, and κ is compared lexicographically. Person missions use $\kappa_c = (\pi_c, b_c, s_c)$ because their relations are not scored and lying-person boxes can distort inferred footprints. For open-space commands without a named landmark, the target maximizing clearance from landmarks is selected unless the best candidates are ambiguous. For an on command with no detected target on the anchor, a local low-threshold search is performed.

B. Situation Assessment and Geometric Localization

Situation assessment applies the organizer's posture–support rules [1] to four posture groups (from detector labels) and five support groups, after an incapacitation override. Support is inferred from the landmark for on/above relations and otherwise from the area cue, so identical lying postures can imply different situations on grass and in transit areas.

For image point (u, v), intrinsic matrix K, camera-to-world rotation R_{wc}, and camera center C, the viewing ray intersects the support plane $z = h$ [4]:

$$d_w = R_{wc}\, K^{-1}\, [u, v, 1]^T, \quad X_w = C + (h - C_z) \,/\, (d_w)_z \cdot d_w \tag{5}$$

The default point is the box bottom center $(u, v) = ((x_{min} + x_{max})/2, y_{max})$, which approximately lies on the camera-side edge of the object. It is pushed away from the camera by the footprint radius ρ, half the projected width of the box bottom edge:

$$\hat{X} = X_w + \rho\, e, \quad e = [(X_w - C)_x, (X_w - C)_y, 0]^T \,/\, \|\cdot\| \tag{6}$$

Only the image and the platform-provided ground height h are used, without asset dimensions or maps. The footprint is an OBB aligned with the viewing ray whose lateral half-length is ρ; because depth is unobservable from one view, its depth half-length is also ρ (square footprint). Targets supported by an anchor are projected onto its top plane, whose height is estimated per object, without class constants, where the ray through the box top center meets the vertical line at the footprint's far edge. Relations use XY footprint geometry: beside/near depend on oriented-box surface separation, while on/above additionally use vertical arrangement. For person targets, ankle keypoints (u_a, v_a) replace the box point:

$$\begin{cases} (u, v, \rho) = (u_a, v_a, 0) & \text{prostrate postures} \\ (u, v, \rho) = (u_a, y_{max}, \rho) & \text{otherwise} \end{cases} \tag{7}$$

with keypoints from a COCO-pretrained model; for prostrate postures the box bottom may be a nearer body part. Standing people keep the box-bottom row because vertical errors grow at shallow viewing angles.

IV. STAGE 2: COST-TO-GO-GUIDED MPPI WITH A DEPTH-PRIORITIZED OCCUPANCY GRID

A. Cost-to-Go-Guided MPPI

At each update a 0.2 m grid combines the static map and sensed obstacles. Each cell has a per-length cost q(x)

$$q(x) = 1 + w_i\, e^{-\alpha(d(x) - r_{in})} + w_u\, u(x) + w_s\, s(x) \tag{8}$$

where d(x) is the distance to the nearest impassable cell, r_{in} = 0.3 m the robot inscribed radius, u(x) the off-road indicator, and s(x) the occupancy soft cost of (16); cells with $d(x) < r_{in}$ are impassable. Dijkstra search from the goal x_g over the 8-neighborhood $\mathcal{N}$ gives the cost-to-go field [7]

$$V(x) = \min_{x' \in \mathcal{N}(x)} [\, V(x') + \bar{q}(x, x')\, \|x - x'\| \,] \tag{9}$$

with $\bar{q}$ the mean cost of the two cells and $V(x_g) = 0$. V has no spurious local minima on the grid, which gives MPPI a consistent descent direction but does not by itself guarantee arrival under motion constraints. MPPI [5], [8] samples M = 1,024 control sequences $u_t = (v_t, \omega_t)$ over T = 25 steps (Δt = 0.1 s) and scores each rollout by

$$J = \Sigma_t [w_p V_t + w_\varphi \varphi_t + w_h(1 - \cos \Delta\theta_t)]\Delta t + w_g V_T + w_c N_{col} \tag{10}$$

where $V_t = V(x_t)$, φ_t is the obstacle-proximity cost, $\Delta\theta_t = \theta_t - \theta_{\nabla V}$ the heading error from the descent direction of V, and N_{col} the number of colliding steps (smoothness and reverse terms omitted). The nominal sequence is updated with weighted perturbations ε_m:

$$w_m \propto \exp(\,-(J_m - J_{min}) \,/\, \lambda\,), \quad u_t \leftarrow u_t + \Sigma_m\, w_m\, \varepsilon_{m,t} \tag{11}$$

Inside each rollout the commands are projected onto the chassis limits

$$|v| \le 1.1 \text{ m/s}, \quad |v\,\omega| \le a_{lat}, \quad |\omega| \le \min(\,1.2,\; 1.8\,|v| \,/\, b\,) \tag{12}$$

with lateral acceleration a_{lat} = 1.0 m/s² and track width b; the steering bound is lifted for pure spins. Structured turning and spin-then-go candidates supplement time-correlated random samples to handle large heading changes. Curb and step lines extracted from ground-height discontinuities join the rollout collision check, and speed is lowered near them.

B. Depth-Prioritized Occupancy Grid and Grasping

Blocking every LiDAR-hit cell lets transient returns accumulate until the goal is enclosed, so each 0.1 m cell keeps a log-odds value L [9], [10]:

$$L(x) \leftarrow \mathrm{clip}(\, L(x) + \ell,\, L_{min}, L_{max}\,), \quad \ell \in \{\, \ell_{occ}, \ell_{free} \,\} \tag{13}$$

LiDAR hits add ℓ_{occ} = 0.85 and traversed cells add ℓ_{free} = −0.4, with $[L_{min}, L_{max}]$ = [−2, 3.5] following OctoMap settings [11]. A beam of range R_b that reaches the ground from sensor height h_s gives free-space evidence only while it is above the obstacle band z_0 = 0.15 m:

$$R_{free} = R_b\,(1 - z_0 / h_s) \tag{14}$$

Because slope-induced ground errors recur in the same cells, repeated observations entrench them instead of averaging them out. Cells in the set F_d that the depth camera confirms as floor therefore take precedence:

$$L(x) \leftarrow L_{min} \quad \text{for } x \in F_d \tag{15}$$

F_d is rebuilt per depth frame: a plane $z = p_x x + p_y y + p_0$ is least-squares fitted to the low points within 4 m and accepted for $|p_x|$, $|p_y| \le 0.27$ (about 15° along each axis) so that slopes are followed; otherwise the 5th height percentile serves as a flat reference. Pixels within 0.05 m of the plane are floor, and those 0.15–1.8 m above it are obstacles. To keep real obstacles from being erased, a frame is skipped when floor covers less than 30% of valid pixels or the downward view is mostly blocked, floor beyond a nearer obstacle in the same 2° bearing is discarded as occlusion shadow, and cells that contain an obstacle point in the same frame are not reset.

The 0.2 m planning grid samples L at each cell center instead of taking the maximum over the cell, so small obstacles are not inflated and narrow gaps stay open. Cells with $L \ge L_h$ = 2.5 are impassable; weaker evidence enters q(x) in (8) as a soft cost dilated by the inscribed radius (L_s = 0.4), so uncertain cells never block the goal:

$$s(x) = \max_{\|x' - x\| \le r_{in}} \mathrm{clip}(\, (L(x') - L_s) \,/\, (L_h - L_s),\, 0,\, 1\,) \tag{16}$$

For grasping, depth points above the floor are clustered in the arm frame, and the cluster nearest the navigation estimate is selected. Upright objects use a top-center grasp; lying objects use their principal-axis geometry to orient the gripper. Detected curbs are excluded from grasp candidates, and up to three feedback-triggered retries are allowed.

V. EXPERIMENTAL RESULTS AND DISCUSSION

A. Stage 1 Evaluation and Cumulative Comparisons

Stage 1 uses all 33 public Chungmu missions (14 person and 19 lost-item missions), with seven network-disabled live runs per configuration. Semantic fields use exact matching and coordinates use XY error. Tables I–II report mean ± standard deviation. The detector obtains mAP50 = 0.841 on 500 platform-generated validation scenes; tiling raises small-object recall from 0.42–0.50 to 0.58–0.71.

The final configuration selects the correct camera in 29/33 rounds in every run, with all target-type or situation-category outputs correct. Its mean total is 75.8/100; the four camera failures involve targets missed in the correct view. Table I shows the largest incremental camera gain after area gating. These cumulative comparisons measure improvements conditional on preceding components, rather than isolated effects.

TABLE I. CUMULATIVE CAMERA-SELECTION RESULTS (7 RUNS, MEAN ± STD)

Configuration	Camera (%)	Score /100
G0 Confidence only	33.8 ± 1.1	44.8 ± 0.5
G1 + Landmark gate (1)	40.7 ± 1.6	49.1 ± 1.0
G2 + Area gate (2)	75.8 ± 0.0	70.1 ± 1.8
G3 + Relation ranking (3)–(4)	84.8 ± 0.0	75.5 ± 1.7
G4 + Person proximity (final)	87.9 ± 0.0	75.8 ± 0.6

Table II compares localization stages. Footprint correction reduces anchor error from 1.74 m to 0.64 m in correctly selected object rounds. Support-height correction then yields the largest total-score increase in this sequence. Local re-detection improves on-command scores, and ankle correction reduces person error from 0.93 m to 0.75 m. Small score differences are not treated as evidence of statistical significance. About half of zero-target-score rounds stem from targets missed in the correct camera, a limitation that geometric correction alone does not resolve.

TABLE II. CUMULATIVE LOCALIZATION RESULTS (7 RUNS, MEAN ± STD)

Configuration	Score	On score	Person (m)
C0 Box center	59.9 ± 1.7	37.5 ± 0.0	2.00 ± 0.01
C1 Box bottom (5)	64.7 ± 0.8	47.3 ± 0.0	1.49 ± 0.00
C2 + Footprint (6)	63.2 ± 1.0	43.7 ± 0.0	1.46 ± 0.00
C3 + Support height	72.3 ± 1.4	68.1 ± 0.0	0.93 ± 0.00
C4 + Re-detection	73.8 ± 1.4	75.5 ± 1.9	0.93 ± 0.00
C5 + Ankles (7) (final)	75.8 ± 0.6	76.0 ± 3.4	0.75 ± 0.00

On score averages the 11 on commands. Person error is the per-run median over the 14 person missions, averaged across runs. C4 changes detection, not only geometry.

B. Stage 2 Retrieval and Obstacle Representation

Five public missions give natural-language retrieval commands for tumblers and umbrellas, two of them among dense obstacles around a taxi. Pick (50) and delivery (30) points form the retrieval subtotal. A run counts as completed when the delivery points are awarded, regardless of grasp correctness. A run is stopped with zero score when a watchdog window shows neither a task milestone nor a 0.3 m gain in straight-line goal distance, extended while recovery actions are active. Table III compares the initial implementation (A* + proportional control + arc detour) with the final system over four rounds per mission; its middle column (MPPI on the initial disc obstacles) separates the contributions of cost-to-go-guided MPPI and the depth-prioritized grid.

TABLE III. RETRIEVAL RESULTS ON FIVE MISSIONS (4 ROUNDS, 20 RUNS EACH)

Metric	A* + P control (discs)	MPPI (discs)	MPPI (depth grid)
Completed / runs	9/20	18/20	20/20
Retrieval score /80	36.0 ± 40.8	69.5 ± 26.3	75.0 ± 15.4
Stage 2 score /100	42.8 ± 48.7	82.5 ± 30.2	90.1 ± 14.5
Correct grasps[1]	9/20	17/20	18/20
Drive time (s)[2]	107.1	117.9	85.0
Brakes per run	15.0	4.6	1.0
Path-loss runs	–	7	0

[1] Correct grasps confirmed by the platform, which scores only completed runs. [2] Mean of per-mission means over missions completed by all three configurations. Brakes: LiDAR proximity stops. Path loss: V = ∞ at the robot in some field build.

With the same disc obstacles, cost-to-go-guided MPPI raised completion from 9/20 to 18/20 and cut brakes from 15.0 to 4.6 per run; the initial implementation mostly stalled after losing its A* path, whereas the cost-to-go field kept a descent direction. The depth-prioritized grid removed the remaining path loss (7 runs), completing all 20 runs with a common-mission drive time of 85.0 s and 1.0 brake per run.

With MPPI fixed in the dense area (Table IV), discs without pass-through clearing produced path loss in 5/8 runs. A conventional occupancy grid failed in both tested runs as slope-induced errors accumulated. The occupancy grid combined with depth priority completed all four tumbler and four umbrella trials in means of 107.2 s and 110.8 s, without path loss or relaxed field builds.

TABLE IV. DENSE-AREA OBSTACLE-REPRESENTATION RESULTS (MPPI)

Representation	Completed[1]	Path lost	Relaxed builds[2]	Time (s)[3]
Discs	3/4; 3/4	5/8	114.4	145.4; 172.8
Occ. grid (13)–(14)	0/1; 0/1	2/2	70.0	—
+ Depth priority (15)	4/4; 4/4	0/8	0	107.2; 110.8

[1] Tumbler; umbrella. [2] Field builds per run with relaxed obstacle constraints. [3] Means for completed tumbler; umbrella trials.

The remaining grasp failures (2 of 4 final runs) occur with the vending-machine umbrella, where curb fragments remain in the depth window.

VI. FUTURE WORK

The parser and detector use fixed vocabularies. A local language-model extension with Grounding DINO [12] for open-vocabulary re-detection was disabled because limited VRAM prevented a reliable evaluation; we will evaluate it with sufficient memory and improve detection recall.

VII. CONCLUSION

We presented an offline MARC 2026 agent built on relational, geometric grounding and cost-to-go-guided MPPI. Area gating and support-height correction gave the largest Stage 1 gains. In Stage 2 the final configuration raised completion from 9/20 to 20/20 runs: cost-to-go-guided MPPI provided most of the gain (18/20 on the same obstacle map), and depth priority removed the remaining path loss. Curb-related grasp ambiguity remains the main failure source.

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